

A machine-checked proof of an Amos-type bound for modified Bessel ratios at real order

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Abstract

The Amos-type upper bound for the modified Bessel function ratio,

$$\rho_\nu(x) = \frac{I_{\nu+1}(x)}{I_\nu(x)} < \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2}} = B_\nu(x),$$

is classical, and its derivation through the qualitative theory of the associated Riccati equation is an established technique. A companion paper formalized the bound at integer order over a factorial power series. This paper extends the formalization to *every real order* $\nu \geq 0$: the function I_ν is defined by its Γ -power series (real exponents via `rpow`), and the complete chain — convergence, positivity, the three-term recurrence, termwise differentiation with a dominated-derivative argument that must treat the leading term separately (its exponent $\nu - 1$ is negative for $\nu < 1$), the Riccati equation, a small-argument zone bound uniform in ν , and a first-crossing barrier — is machine-checked in Lean 4 with axiom oracle `[propext, Classical.choice, Quot.sound]` and no analytic hypothesis beyond $\nu \geq 0$, $x > 0$. Two structural locks tie the result to the integer development: an *identification theorem* proves that at $\nu = n \in \mathbb{N}$ the Γ -series object coincides with the factorial-series object, so the integer-order theorem of the companion development is recovered as a corollary in three rewrites; and a genuinely non-integer instance at $\nu = \frac{1}{2}$ witnesses that the endpoint lives outside the natural-number embedding. The theorem is proved for the in-core Γ -series definition; no identification with an external special-functions library object is claimed.

1 Introduction

The ratio $\rho_\nu = I_{\nu+1}/I_\nu$ of modified Bessel functions of the first kind is bounded above by the algebraically calibrated expression $B_\nu(x) = x/(\nu + \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2})$; bounds of this shape trace back to Amos [1], and sharp forms and systematic families were developed by Segura [2], Hornik and Grün [3, 5], and Ruiz-Antolín and Segura [4], the latter describing explicitly the Riccati-based qualitative methodology that classical proofs of such bounds employ. *No novelty is claimed for the bound or for the Riccati approach.* We refer to the inequality as an *Amos-type* bound: the attribution of this exact display to a specific equation of [1] is deliberately not asserted here pending collation with the primary source.

*This manuscript and parts of the formal development were produced with AI assistance under the repository’s audit protocol. Every theorem below is machine-checked in Lean 4 against a pinned Mathlib; the independent numerical cross-check of Section 6 is a committed interval-arithmetic transcript; remaining numerics are labelled paper-level where they occur. Repository: <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>.

The contribution of this paper is the formalization at *real order*. Two companion papers of this series formalized, over the factorial power series at integer order, the consequence calculus of the bound [7] and then the bound itself [8]. The latter states its own scope limit: no extension to noninteger order is claimed. This paper removes that limit. The bound and its consumers below are stated for every real $\nu \geq 0$ and every $x > 0$, over the Γ -power series

$$\mathbf{besselIReal} \ \nu \ x = \sum_{k \geq 0} \frac{(x/2)^{\nu+2k}}{k! \Gamma(\nu + k + 1)},$$

with the real exponent carried by `Real.rpow` and the denominator by `Real.Gamma`. In addition, the *identification theorem* `besselIReal_natCast` proves `besselIReal n = besselI n` at every integer order — so the integer development of [7, 8] is not a parallel theory but a proved section of the real one, and the integer-order bound of [8] becomes a corollary (Theorem 3.3).

Statuses: **exact** means a Lean theorem in the pinned central repository with clean axiom oracle; **certified** means a committed interval-arithmetic transcript; **paper-level** means ordinary prose, always flagged; *verified* marks an ordinary high-precision computation used only as a cross-check of a certified result, never as evidence. The repository’s largest namespace (`YangMills/`) reflects its original motivation in lattice gauge theory; no reduction to any continuum or gauge-theoretic problem is claimed or used.

2 Setting

All objects live in module `AmosClosure` of the pinned central repository. From the companion development [7, 8]: the factorial-series `besselI n x =  $\sum_{k \geq 0} (x/2)^{n+2k} / (k! (n+k)!)$` with its positivity, recurrence, derivative and ratio calculus, the predicate `AmosBound  $\nu \ x \ r \iff r < B_\nu(x)$` — already real in its order parameter — and the calibration engine. New in this paper, the real-order object `besselIReal  $\nu \ x$` above, its ratio $\rho_\nu = \mathbf{besselRatioReal} \ \nu \ x$, and $\psi_\nu(x) = x(1/\rho_\nu - \rho_\nu)$ (`besselPsiReal`). Everything is stated at $x > 0$, the `rpow` domain of the development.

3 Main results

Theorem 3.1 (the Amos-type bound at real order; `amosBoundReal_holds`). *For every real $\nu \geq 0$ and every real $x > 0$,*

$$\frac{\mathbf{besselIReal} \ (\nu+1) \ x}{\mathbf{besselIReal} \ \nu \ x} < \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2}}.$$

No hypothesis remains beyond $\nu \geq 0$ and $x > 0$.

Theorem 3.2 (identification; `besselIReal_natCast`). *For every $n \in \mathbb{N}$ and $x > 0$,*

$$\mathbf{besselIReal} \ (n : \mathbb{R}) \ x = \mathbf{besselI} \ n \ x.$$

Termwise, `rpow` collapses to the natural power and $\Gamma(n + k + 1) = (n + k)!$. The integer-order development is a proved restriction of the real-order one.

Theorem 3.3 (consumers, recovery, and the half-order witness). *For every real $\nu \geq 0$ and $x > 0$: $\rho_\nu(x) - \rho_{\nu+1}(x) < 1/x$ (`besselIReal_unit_step`); the derivative of $\log \circ \text{besselIReal } \nu$ at x is strictly below that of $\log \circ \text{besselIReal } (\nu+1)$, in `deriv` form (`besselIReal_logDeriv_lt`). Moreover the integer-order theorem of [8] is recovered through Theorem 3.2 in three rewrites (`amosBound_holds_recovered`), and the instance $\nu = \frac{1}{2}$ is stated and proved as a named theorem (`amosBoundReal_half_order`) — the endpoint is not consumed only through the natural-number embedding.*

The φ -monotonicity step of [7] remains integer-indexed — its sequence is integer by construction and is already unconditional through [8]; no artificial real-order variant is introduced.

4 The proof

The formal proofs are the authority; this section presents the mathematics in full, at the level of the checklist a referee needs.

4.1 Series facts

Write $t_{\nu,k}(x) = (x/2)^{\nu+2k} / (k! \Gamma(\nu + k + 1))$. The exact term-ratio identity

$$t_{\nu,k+1} = t_{\nu,k} \cdot \frac{(x/2)^2}{(k+1)(\nu+k+1)} \quad (\text{besselRealTerm_succ})$$

follows from $\Gamma(s+1) = s \Gamma(s)$ at $s = \nu + k + 1 \geq 1$ and $(x/2)^{a+2} = (x/2)^a (x/2)^2$. Convergence: since $\Gamma(\nu+1) \leq \Gamma(\nu+k+1)$ for $\nu \geq 0$ (each functional-equation factor $\nu+j \geq 1$), the terms are dominated by an exponential series, and $I_\nu > 0$ follows termwise. The three-term recurrence $I_\nu - I_{\nu+2} = (2(\nu+1)/x) I_{\nu+1}$ is proved termwise (index shift plus the functional equation), exactly as at integer order. Theorem 3.2 is likewise termwise: $(x/2)^{(n:\mathbb{R})+2k} = (x/2)^{n+2k}$ by `Real.rpow_natCast` and $\Gamma(n+k+1) = (n+k)!$ by `Real.Gamma_nat_eq_factorial`.

4.2 The derivative layer

Termwise differentiation on the interval $(x/2, 3x/2) \subset (0, \infty)$ — the `rpow` domain forbids a ball through 0 — with dominated derivatives. The termwise derivative is

$$\frac{d}{dy} t_{\nu,k}(y) = \frac{\nu+2k}{2} \frac{(y/2)^{\nu+2k-1}}{k! \Gamma(\nu+k+1)}.$$

Remark 4.1 (the negative-exponent trap). *For $k = 0$ and $0 \leq \nu < 1$ the exponent $\nu - 1$ is negative: the factor $(y/2)^{\nu-1}$ is decreasing in y , and a majorant that evaluates it at the right endpoint is false. The formalized domination therefore treats $k = 0$ separately, with the two-endpoint bound $(y/2)^{\nu-1} \leq (x/4)^{\nu-1} + (3x/4)^{\nu-1}$ valid in both monotonicity regimes (at $\nu = 0$ the whole term carries the factor $\nu = 0$); and for $k \geq 1$ it uses the factoring $\nu+2k-1 = (\nu+1) + 2(k-1)$, whose exponents stay ≥ 1 , giving $(y/2)^{\nu+2k-1} \leq (3x/4)^{\nu+1} A^{k-1}$ with $A = (3x/4)^2$. The majorant tail is summable since $(\nu+2k+2)/(k+1)! \leq (\nu+2)/k!$, equivalent to $0 \leq \nu k$.*

Mathlib's dominated-differentiation theorem for series then yields, for every $\nu \geq 0$ and $x > 0$,

$$I'_\nu(x) = I_{\nu+1}(x) + \frac{\nu}{x} I_\nu(x) \quad (\text{besselIReal_hasDerivAt}),$$

with the `deriv`-form and logarithmic-derivative identities as corollaries, and the $\nu = n$ value test against the integer layer through Theorem 3.2 (`besselIReal_deriv_value_natCast`).

4.3 The Riccati equation and the nullcline

From the derivative identity and the recurrence, the quotient rule gives

$$\rho'_\nu(x) = Q_x(\rho_\nu(x)), \quad Q_x(y) := 1 - \frac{2\nu+1}{x} y - y^2 \quad (\text{besselRatioReal_hasDerivAt}). \quad (1)$$

The bound B_ν satisfies the exact calibration identity

$$\frac{1}{B_\nu(x)} - B_\nu(x) = \frac{2\nu+1}{x} \quad (\text{amosRHS_calibration_real}), \quad (2)$$

with $a = \nu + \frac{1}{2} > 0$ supplied by $\nu \geq 0$; multiplied by $B_\nu > 0$ this is precisely $Q_x(B_\nu(x)) = 0$ (`riccatiQReal_amosRHS`): the bound is the positive root of the Riccati quadratic. Moreover $Q_x(y) > 0$ for $0 \leq y < B_\nu(x)$, by the factorization $Q_x(y) - Q_x(B_\nu) = (B_\nu - y)(\frac{2\nu+1}{x} + y + B_\nu)$ (`riccatiQReal_pos_of_lt`).

4.4 The zone: $x \leq 1/4$, uniformly in ν

The target $\rho_\nu < B_\nu$ is equivalent, by (2) and the strict antitonicity of $t \mapsto 1/t - t$ on $(0, \infty)$, to $\psi_\nu > 2\nu + 1$. On the zone this is proved directly from the series, with four product-form facts: geometric domination $t_{\nu,k} \leq t_{\nu,0} q_\nu^k$ with $q_\nu = (x/2)^2/(\nu+1)$, by induction from the term-ratio identity and $\nu+1 \leq (k+1)(\nu+k+1)$ (`besselRealTerm_le_geometric`); the product bound $(1 - q_\nu) I_\nu \leq t_{\nu,0}$ for $q_\nu < 1$ (`besselIReal_mul_le`); the strict two-term lower bound $t_{\nu,0} < I_\nu$ (`besselRealTerm_zero_lt_besselIReal`) — the sole source of strictness in the whole chain; and the exact identity $2(\nu+1) t_{\nu+1,0} = x t_{\nu,0}$, derived by multiplying the already-proved termwise recurrence at $k = 0$ by x (`besselRealTerm_zero_succ`).

The ν -uniformity is isolated in two named lemmas rather than automation: $(\nu+1)/(\nu+2) \leq 1$ (`real_zone_ratio_uniform`), whence $2(\nu+1) q_{\nu+1} = \frac{x^2}{2} \cdot \frac{\nu+1}{\nu+2} \leq \frac{x^2}{2}$, and with $x^2 \leq 1/16$ the coefficient inequality

$$2\nu + 1 + x^2 \leq 2(\nu+1)(1 - q_{\nu+1}) \quad (\text{real_zone_coefficient_bound}).$$

Chaining (upper bound at order $\nu+1$, identity, strict lower bound at order ν) gives the decisive strict inequality

$$(2\nu + 1 + x^2) I_{\nu+1}(x) < x I_\nu(x) \quad (0 < x \leq 1/4), \quad (3)$$

strictness entering only at the last link. Two further lines convert (3) into the zone claim: since $2\nu+1+x^2 \geq 1$, (3) also gives the crude bound $I_{\nu+1} \leq x I_\nu$, hence $x I_{\nu+1}^2 \leq x^2 I_{\nu+1} I_\nu$; multiplying (3) by $I_\nu > 0$ and inserting this,

$$(2\nu + 1) I_{\nu+1} I_\nu + x I_{\nu+1}^2 \leq (2\nu + 1) I_{\nu+1} I_\nu + x^2 I_{\nu+1} I_\nu < x I_\nu^2,$$

which, by the identity $\psi_\nu \cdot (I_{\nu+1} I_\nu) = x(I_\nu^2 - I_{\nu+1}^2)$, is precisely $\psi_\nu(x) > 2\nu + 1$ on the zone (`besselPsiReal_zone`).

4.5 The barrier: every touch is an upward crossing

Suppose $\psi_\nu(c) = 2\nu + 1$ at some $c > 0$. Then $1/\rho_\nu(c) - \rho_\nu(c) = (2\nu + 1)/c$, and substituting into the Riccati quadratic annihilates it, so $\rho'_\nu(c) = 0$ (`riccati_zero_of_real_touch`); the product rule collapses $\psi'_\nu(c) = (2\nu + 1)/c > 0$, the numerator positive directly from $\nu \geq 0$. Every touch of the critical level is an upward crossing.

Lemma 4.2 (first crossing; the human form of `besselPsiReal_gt`). *Suppose, for contradiction, $\psi_\nu(x_1) \leq 2\nu + 1$ for some $x_1 > 0$. By the zone theorem $x_1 > 1/4$. The set $S = \{y \in [1/4, x_1] : \psi_\nu(y) \leq 2\nu + 1\}$ is nonempty, bounded below, and closed (continuity of ψ_ν on the interval, from differentiability); let $c = \inf S \in S$. Since $\psi_\nu(1/4) > 2\nu + 1$, in fact $c > 1/4$, and by minimality $\psi_\nu > 2\nu + 1$ on $(1/4, c)$ (at $1/4$ itself this holds by the zone theorem); left-continuity forces the touch $\psi_\nu(c) = 2\nu + 1$. But then $\psi'_\nu(c) = (2\nu + 1)/c > 0$, so the difference quotient $(\psi_\nu(y) - \psi_\nu(c))/(y - c)$ is eventually positive as $y \uparrow c$; since $y - c < 0$ there, $\psi_\nu(y) < \psi_\nu(c) = 2\nu + 1$ just left of c — contradicting $\psi_\nu > 2\nu + 1$ on $(1/4, c)$. Hence $\psi_\nu(x) > 2\nu + 1$ for all $x > 0$.*

Finally, $\psi_\nu > 2\nu + 1$ converts into $\rho_\nu < B_\nu$ algebraically: with $s = 1/\rho_\nu - \rho_\nu$ and $t = 1/B_\nu - B_\nu$,

$$(s - t) \cdot \rho_\nu B_\nu = (B_\nu - \rho_\nu)(1 + \rho_\nu B_\nu),$$

and the left-hand side is positive — both its factors are: $s > t$ is the barrier conclusion $\psi_\nu > 2\nu + 1$ divided by x , and $\rho_\nu B_\nu > 0$ — so, since $1 + \rho_\nu B_\nu > 0$, positivity passes to $B_\nu - \rho_\nu$; this is the closing step of `amosBoundReal_holds`. That completes Theorem 3.1; Theorem 3.3 is a direct application, the recovery corollary consisting of the two identification rewrites, one cast normalization, and `exact`.

5 Scope

The theorem is proved for the in-core Γ -power-series definition of the modified Bessel function at real order $\nu \geq 0$, $x > 0$. No formal identification with an external special-functions library object is claimed (none exists at the pinned Mathlib to identify with). The integer-order objects of the companion development are identified with the real-order object *by theorem* (Theorem 3.2) — the corresponding scope caveat of [8] is thereby discharged. Results elsewhere in this programme that consume Amos-type bounds for the classical I_ν of the analysis literature — such as the surface-track φ -lemma conditional discussed in [7] — do not change verification status through this paper.

6 Independent numerical cross-check

A certified interval-arithmetic companion evaluates the margin — its protocol pre-registered in the development's charter (`docs/C4-CHARTER.md`) before the authoritative run — $\Delta(\nu, x) = B_\nu(x) - \rho_\nu(x)$ at all 70 points of the pre-registered grid $\nu \in \{0, \frac{1}{10}, \frac{1}{2}, \frac{9}{10}, 1, \frac{3}{2}, 2, \pi, 10, 100\}$, $x \in \{\frac{1}{8}, \frac{1}{4}, 1, 2, 5, 50, 300\}$, with π a certified ball. The ratio is computed by the certified backward recurrence $\rho_\alpha = x/(2(\alpha + 1) + x\rho_{\alpha+1})$, anchored at order $\nu + N$ with the anchor condition $q \leq \frac{1}{4}$ itself certified in ball arithmetic and seeded by the two-sided product-form bounds of Section 4.4 — no exponentially large value of I_ν is ever formed, and the seed is independent of Theorem 3.1. Verdicts are by ball sign (PASS iff $\inf \Delta > 0$), never by midpoint. All 70 points PASS at 128 bits; an independent ordinary high-precision pass (labelled *verified*, not certified) agrees at all

70 points; and at $(\nu, x) = (0, 300)$ the certified margin $[2.78243072419 \times 10^{-6} \pm 2.43 \times 10^{-19}]$ reproduces, by this entirely different method, the slack floor measured by the integer-order companion of [7]. It is *not* part of the proof, and the proof does not cite it.

7 Related work

Amos [1] introduced bounds of this shape in the context of computing modified Bessel ratios; two-sided sharp families and Turán-type consequences are due to Segura [2]; Hornik–Grün studied the Amos-type family systematically [3] and proved in-family optimality results [5]; Ruiz-Antolín–Segura [4] give the sharp bounds at both parameter ends and describe the Riccati-based qualitative methodology to which the route formalized here belongs. The recurrence and derivative identities are [6, 10.29], and the Γ -series is the standard definition [6, 10.25.2]. On the formalization side we know of no prior machine-checked proof of an Amos-type ratio bound at real order; the integer-order formalization and its consequence calculus are in the companion papers [7, 8]; the prose companions of the consumer families are [9, 10, 11]. The development builds on Mathlib [12].

8 Reproducibility

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit

07642720480157414db592fa85b626dafb71355b

(lakefile and manifest agree); `tectonic` 0.15.0; `python-flint` 0.9.0 for the certified companion.

Recorded build facts: at the final mathematics commit

56ff479b863d8585c41f6738b7bd18f80f2cbb16

`lake build AmosClosure` completed successfully (**8171 jobs**); the axiom oracle printed, for all **74** registered statements (the 34 of the `c3-v1.0` release registry plus the 40 of this paper), exactly

`[proptext, Classical.choice, Quot.sound];`

the oracle log is committed under `scripts/` at the same commit. The full run history of this development — eight build transcripts including the vacuous first run and every failed attempt, each committed the day it was produced with its diagnosis in the commit message; the certified-grid transcripts (the first kept but superseded; the second the authoritative script-and-transcript pair); and six charter amendments recording every external audit round and every method deviation — is committed under `scripts/` and `docs/C4-CHARTER.md`; per the repository’s audit protocol it is archival material and plays no role in the mathematics above.

Headline-theorem-to-artifact map (all in module `AmosClosure`; every row carries a direct oracle witness; the 34 inherited statements are mapped in [7, 8]):

Result	Lean name	File	Status
Γ -series interface	summable_besselRealTerm etc.	BesselRealInterface.lean	exact
Thm. 3.2	besselIReal_natCast	BesselRealInterface.lean	exact
derivative identity	besselIReal_hasDerivAt	BesselRealDeriv.lean	exact
Eq. (1)	besselRatioReal_hasDerivAt	RiccatiReal.lean	exact
Eq. (2)	amosRHS_calibration_real	RiccatiReal.lean	exact
zone (§4.4)	besselPsiReal_zone	AmosBoundProofReal.lean	exact
uniformity lemmas	real_zone_ratio_uniform etc.	AmosBoundProofReal.lean	exact
Lemma 4.2	besselPsiReal_gt	AmosBarrierReal.lean	exact
Thm. 3.1	amosBoundReal_holds	AmosBarrierReal.lean	exact
consumers	besselIReal_unit_step, _logDeriv_lt	AmosBarrierReal.lean	exact
recovery lock	amosBound_holds_recovered	AmosBarrierReal.lean	exact
half-order witness	amosBoundReal_half_order	AmosBarrierReal.lean	exact
§6 grid	c4_amos_real_arb.py + transcript v2	scripts/	certified

From a clean clone at the release revision:

```
lake exe cache get
lake build AmosClosure
lake env lean AmosClosure/Oracle.lean
python scripts/c4_amos_real_arb.py
tectonic -X compile papers/c4-amos-real/c4_amos_real.tex
```

At the release revision, canonical artifact hashes will be recorded — tag-scoped, in `RELEASE-MANIFEST.md` in this paper’s directory — committed together with the release revision of this manuscript; compare against `git show <rev>:path`, never a worktree checkout. The annotated tag `c4-v1.0` will mark that release commit after the pre-release audits close, with the audit registry in `docs/C4-CHARTER.md`; the release-tree audit is recorded in the manifest, post-tag by sequencing, with any fixes in post-tag commits; tags never move.

9 Conclusion

An Amos-type bound for modified Bessel ratios is now a theorem of the pinned formal development at every real order $\nu \geq 0$, proved from the in-core Γ -power series using only machine-checked analytic library results, with no additional analytic hypothesis beyond $u \geq 0$ and $x > 0$, and with the integer-order development of the companion papers identified as a proved restriction and recovered as a corollary. Limitations: the in-core series definition only, and $\nu \geq 0$ (Section 5); the classical bound and the Riccati methodology are prior art — the contribution is the first machine-checked formalization of this complete chain in the repository, and we are not aware of a prior machine-checked formalization of the result. Natural next steps: the sharp lower bound (whose small-argument analysis is second-order, as any lower bound matches the first Taylor coefficient of ρ_ν), formal identification with a general special-functions interface if one lands in Mathlib, and upstreaming the modified-Bessel API.

References

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